ENGINEERING RESEARCH REPORT

Fringe-Projection Depth Cameras: Failure Modes, Self-Calibration, and Noise

A mechanism-first guide to phase aliases, optical ghosts, exposure failures, plane calibration, and heteroscedastic triangulation noise.

Audience: Engineers diagnosing or characterizing an undocumented fringe-projection or triangulation depth unit

Date: 4 September 2026

Scope: Single-shot and few-shot fringe/phase-shift structured light; phase-order aliases; optical ghosts; exposure; plane-based calibration; and per-point noise. The report is intentionally product-agnostic.

Core result	A one-period phase-order mistake is normally a constant shift in disparity or reciprocal depth - not a constant fraction of true range. A fixed phantom sheet is therefore more suggestive of a stable stray-light phase field or a vendor-specific lookup behavior unless a distance sweep follows the predicted alias family.

Prepared from peer-reviewed optics and calibration literature, NIST metrology guidance, and current first-party scanner documentation. All source links are clickable.

Direct answer

The five phenomena in the question should be treated as different layers of one measurement chain:

1. **Phase-order aliasing is a correspondence error.** A sinusoidal decoder recovers phase modulo 2 pi. Selecting the wrong fringe order shifts the inferred projector coordinate by an integer number of fringe periods. In rectified geometry this is an integer shift in disparity, so the aliases are evenly spaced in **inverse depth**, not normally at a fixed percentage of true depth.
2. **An internal-reflection ghost is an unwanted light path.** If that path carries a repeatable copy of the projected modulation into the camera, the normal decoder can triangulate it into a stable phantom surface. Its stability is tied to camera/housing coordinates and optical power, not to a fringe-order relation with the real surface.
3. **Exposure errors first damage modulation, then phase, then geometry.** Underexposure, ambient washout, and ordinary clipping reduce the usable sinusoidal component or bias it. Active structured light does not require natural surface texture; the important variables are returned pattern modulation, wavelength-specific reflectance, focus, and specularity.
4. **Flat boards can calibrate a depth system, but only with the right observables and poses.** A printed metric planar target at varied tilts can estimate focal length, principal point, distortion, and pose. Unmarked, fronto-parallel boards placed only at several distances cannot robustly identify all of those parameters; they are much better for range scale/bias and field-error maps.
5. **Valid-point precision is heteroscedastic and outliers are a separate population.** For triangulation, axial standard deviation contains the familiar z-squared geometric amplification. Phase noise also grows when modulation falls because of distance, low reflectance, grazing illumination, defocus, or ambient light. Aliases, multipath points, and saturation failures should not be forced into a single Gaussian noise law.

The most decisive diagnostic is a controlled matte-board experiment that records raw phase, modulation/amplitude, saturation masks, and final depth while independently sweeping board distance, exposure, projector power, target reflectance, fringe pitch (if available), and firmware thresholds.

1. How fringe projection decodes depth

1.1 Wrapped phase from one or several images

An ideal N-step phase-shift sequence at camera pixel (u,v) is

```
I_n(u,v) = A(u,v) + B(u,v) cos(phi(u,v) + delta_n)
```

where A is the DC or mean intensity, B is fringe modulation amplitude, and delta_n is the commanded phase shift. The usual estimator takes the angle of two weighted sums:

```
phi_wrapped = atan2(sum I_n sin delta_n, sum I_n cos delta_n)
```

The arctangent can return only one 2 pi interval. The absolute phase is therefore

```
Phi = phi_wrapped + 2 pi k
```

where integer k is the fringe order. This formulation and the central role of k are described clearly by Zhang's absolute-phase review and the standard phase-shifting equations are given in Li et al.'s structured-light calibration paper.

A few-shot system may obtain phi from three or more temporal phase steps. A single-shot system must encode the missing information spatially, spectrally, through a second view, through multiple carrier frequencies in one frame, or through a learned prior. Spatial/Fourier single-shot schemes trade spatial bandwidth and local continuity assumptions for temporal speed. A single high-frequency sinusoid, by itself, does not contain enough information to identify k on disconnected or depth-discontinuous objects.

Absolute phase can be recovered with multi-frequency fringes, Gray/binary codes, phase codes, geometric constraints, a second camera, or prior scene/CAD knowledge. Every method can fail when the auxiliary cue is noisy, occluded, saturated, outside its geometric search volume, or inconsistent with motion between shots. Zhang 2018 surveys these families and explicitly notes that unwrapping artifacts remain possible because of noise and sampling.

1.2 What a one-period mistake does to depth

After calibration, phase identifies a projector coordinate. If the projector fringe pitch is P pixels, changing k by integer m changes that coordinate by mP. In a rectified camera-projector pair, let disparity be d = u_c - u_p and let F = fB_g be camera focal length in pixels times the physical camera-projector baseline. Then

```
z_true = F / d
```

and a wrong order gives

```
z_m = F / (d - mP)
```

or, equivalently,

```
1/z_m = 1/z_true - mP/F
```

The sign changes if the disparity convention is reversed. The invariant result is that adjacent aliases are separated by a constant increment in **disparity or reciprocal depth**, not by a constant increment in depth and not by a constant ratio z_m/z_true. The ratio is

```
z_m/z_true = 1 / (1 - mP z_true/F)
```

so it varies with true depth and can become very large near zero disparity. This is the same inverse-depth structure measured for a triangulation depth sensor by Khoshelham and Oude Elberink.

For a general non-rectified pair, each candidate fringe order selects a different projector ray or phase plane; the alias is the intersection of that candidate with the camera ray. There is no global fixed fraction or globally constant axial spacing. A local small-angle model can look simpler: Jiang et al. derive an approximate single-period unambiguous depth interval Delta z = T_s/sin(alpha), where T_s is the projected spatial fringe period and alpha is the camera-projector angle. Over a small field and shallow volume, successive aliases can therefore resemble roughly parallel slabs separated by an approximately constant axial interval. That is a local approximation, not a universal percentage law.

An observed alias at a stable fraction of the true range can still occur accidentally over a narrow range, from a vendor's per-pixel phase-to-depth lookup table, from a normalized output convention, or from a second error layered on top of phase wrapping. It should not be assumed from fringe physics alone.

1.3 Out-of-range surfaces and alias signatures

If an absolute-phase method uses a calibrated depth interval to choose k, a surface outside that interval can be assigned to the in-range candidate with the same wrapped phase. Jiang et al. show the limiting case directly: two points on the same camera ray, one period apart in absolute phase, are indistinguishable to the wrapped-phase constraint, and the farther point can be reconstructed at the nearer candidate depth.

The strongest signatures of a decoding alias are:

- The bad point is related to the real point by an integer phase offset: Phi_bad - Phi_true is close to 2 pi m, or the inferred projector coordinate differs by mP.
- Sweeping true board distance makes the phantom follow the projective alias curve above, often with a discontinuous jump when k changes.
- Changing fringe pitch changes the ambiguity interval or reciprocal-depth spacing.
- A z-range or search-volume change may select a different candidate order.
- The location is not fundamentally fixed to a dirty window, lens element, or camera pixel mask.

1.4 What the firmware settings probably mean

These labels are not standardized. The exact definition can be established only from the device documentation, source code, or a controlled sweep. Their conventional meanings are:

Setting	Common implementation	What increasing it usually does	Important ambiguity
Background threshold	Minimum accepted brightness above a dark/background estimate; minimum bright-dark difference; or minimum fringe amplitude/modulation B	Rejects more dim, low-contrast, or shadow pixels	Some vendors call an amplitude/contrast threshold "background" even though it is not the DC level A
Gradient threshold	Maximum permitted local phase, disparity, or depth gradient; less commonly, a minimum image-gradient threshold used to find stripes	For a maximum-gradient control, rejects more edges, unwrap discontinuities, and isolated outliers as the threshold is lowered	Direction is reversed if it is a minimum edge-strength control
Z range	Lower/upper accepted depth in sensor coordinates; sometimes also the candidate search interval used for correspondence or fringe order	Narrows the returned measurement volume	It may be post-reconstruction clipping only, or it may influence ambiguity resolution

A commercial surfaceControl3D integration documents **amplitude threshold** as minimum sine-pattern contrast, **maximum phase gradient** as an outlier rejection control, and upper/lower Z limits as rejection bounds in sensor coordinates (Viper.NET documentation). NIST likewise notes that practical scanners use intensity thresholding to exclude pixels outside preset intensity bounds (Rachakonda et al., NIST). These examples support the conventional interpretations but do not prove the semantics of an unrelated firmware.

Three short tests identify the behavior:

1. Put a matte board fully inside the nominal volume and sweep one setting at a time while retaining raw intensity, modulation, and depth.

2 Move the board outside the configured Z interval. If the points reconstruct unchanged and merely disappear, Z range is a final clip. If the selected depth changes, it participates in correspondence/order search.

3 Present a smooth tilted plane and a sharp depth step. A maximum phase/depth-gradient filter removes the step and isolated points; a minimum image-gradient detector instead removes smooth/weak stripes.

2. Ghost depth from internal reflections

2.1 From a stray optical path to a phantom surface

Optical ghost images are faint secondary images formed by reflections between optical surfaces; they reduce contrast and can refocus on the sensor (Edmund Optics). In a projector-camera depth unit, unwanted paths include:

- projector-to-cover-to-camera coupling inside a common window or housing;
- reflections between camera lens elements, sensor cover glass, filters, and enclosure windows;
- scattering from dust, scratches, adhesive edges, baffles, or a glossy bezel;
- real scene interreflections, particularly in concavities and shiny bins; and
- subsurface scattering or diffusion in translucent materials.

The last two are scene transport rather than internal housing ghosts, but all add light that did not travel along the calibrated projector-surface-camera path. Gupta et al. show that long-range interreflections and short-range subsurface transport can cause large, systematic structured-light decoding errors; ambient illumination is a distinct phenomenon because it can be measured with the projector off.

If the unwanted path contains a modulated copy of the projected pattern, the phase estimator does not know that it arrived by the wrong route. It extracts the phase of the vector sum of direct and stray modulation. If the stray component dominates in a repeatable image region, the calibration then converts that false phase into a repeatable 3D surface. A fixed phantom plane or curved sheet is therefore plausible, but its exact shape is the triangulated image of the stray phase field, not the physical reflecting pane itself.

2.2 Brightness dependence is path-dependent

"The ghost changes with scene brightness" does not identify one mechanism, because several regimes are possible:

- **Direct internal crosstalk:** may remain with a black target or capped external scene and scale mainly with projector power and camera exposure.
- **Return-light lens/window ghost:** strengthens when a bright or specular scene region sends more light back into the receiving optics. Its image may move predictably around the optical axis as the bright source moves.
- **External interreflection:** changes with scene geometry, concavity, material, and illumination, and moves with the scene rather than remaining tied to the housing.
- **Competition between real and fixed components:** on a dark, distant, or oblique target, the real modulation B falls and a small fixed stray component can dominate. On a brighter target, the correct direct signal can overwhelm it. At still higher brightness, lens ghosts or saturation can worsen again. The response can therefore be non-monotonic.

2.3 Distinguishing ghosts from phase aliases

Observation	Phase-order alias	Internal optical ghost	Scene interreflection	Exposure/threshold failure
Relation to true phase	Integer 2 pi / projector-pitch offset	No required integer relation	Biased mixture of phases	Often noisy or harmonically biased, not a clean integer order
Board distance sweep	Follows a projective alias branch; may jump orders	Often stable in camera/housing coordinates	Changes with scene arrangement	Improves or disappears in an exposure window
Fringe-pitch change	Alias spacing changes predictably	Image-space ghost path persists; triangulated shape may change but not by alias law	Error changes with pattern frequency/transport	Depends on modulation and decoder
Projector-off frame	No structured alias	Reveals only ambient/dark/ receiving-optics flare; direct projector crosstalk requires projector on	Ambient component visible; projected interreflection absent	Reveals ambient offset and saturation
Projector-power sweep	Geometry stable until SNR/order failure	Ghost amplitude often scales strongly	Both direct and global components scale	Exposes clipping and low-SNR limits
Rotate/translate target	Alias remains tied to real surface correspondence	Fixed camera-pixel mask is strong evidence	Moves with target/scene	Moves with bright/dark regions
Baffle/window/polarizer change	Little effect unless SNR changes	Can change strongly	Polarization/baffles may help depending on path	Polarization can trade glare for lower signal

The distinction should be made in **raw camera/phase coordinates before point-cloud filtering**. A post-filtered point cloud can make all four mechanisms look like a smooth secondary sheet.

3. Saturation, low signal, and exposure

3.1 How clipping and underexposure corrupt phase

The phase estimator assumes that the sampled intensities are a sinusoid plus noise. Underexposure makes B comparable to read noise, quantization, dark current, and ambient variation. Ordinary overexposure clips some phase steps at the ADC ceiling, converting a sinusoid into a waveform with harmonics and destroying the true relative sample values. Either condition increases phase variance; asymmetric clipping also creates systematic phase bias.

There is a narrow exception worth preserving: Chen and Zhang proved that certain digitally generated, equally spaced N-step sequences can recover accurate phase despite symmetric saturation when N is tied to the integer fringe period and phase steps are exact. Their demonstrated cases used many phase steps, such as N = P/2. That result should not be generalized to an undocumented single-shot or three/four-step implementation, unequal channel responses, rolling exposure, projector gamma, motion, or arbitrary partial clipping. For conventional operation, avoid clipping.

The right exposure target is not merely a pleasing mean image. It is high modulation B with every required phase sample inside the camera's linear response and enough headroom for local highlights. The 2023 precision model of Lv and Qian gives, for N-step phase shifting,

```
variance(phi) = 2(C_noise + K A) / (B^2 N)
```

where K A represents signal-dependent camera noise and C_noise groups approximately signal-independent terms. This captures why a high DC level with weak modulation can still be poor, and why simply adding gain does not recreate lost photons or clipped samples.

3.2 "Low texture" is usually the wrong diagnosis

Passive stereo needs natural texture. Structured light supplies its own. A uniformly painted matte panel can be an excellent target because the projected fringes provide correspondence. What practitioners often call low texture in this context is one of the following:

- low wavelength-specific albedo, producing small B;
- ambient light increasing A without increasing B;
- projector or camera defocus washing out high-frequency fringes;
- a specular surface returning light away from the camera except at clipped highlights;
- translucent or subsurface transport blurring phase spatially;
- a spatial single-shot algorithm losing local carrier bandwidth at a depth edge or steep slope; or
- the surface leaving the overlap of projector and camera fields.

NIST emphasizes that active patterns add texture and simplify correspondence, while dark, shiny, concave, out-of-focus, shadowed, and saturated regions remain difficult (Rachakonda et al.).

3.3 Exposure strategies used in commercial systems

Commercial practice is a layered strategy rather than one magic exposure:

1. **Narrow-band blue illumination and filtering.** The narrow spectral band helps reject broadband ambient light. ZEISS attributes this ambient-interference advantage to blue light (ZEISS blue-light overview). NIST cautions that broad claims that blue intrinsically reduces reflections are not well established for every application.
2. **High optical power and short exposure.** A bright source raises modulation while short exposure suppresses motion and ambient integration. ZEISS describes this combination for ATOS 5, including shiny and dark surfaces (ATOS 5).

3. **Automatic exposure on a relevant ROI.** Use a region that contains the measurement-critical material, not a bright irrelevant background.
4. **Multi-exposure/HDR depth.** Capture complete fringe sequences at several exposure times and, per pixel, use or fuse the strongest unsaturated modulation. NIST reports commercial use of merged scans at multiple exposure, gain, or aperture settings. Mech-Mind's current camera documentation explicitly supports one to three 3D exposure times, with all exposures used for depth and a completeness-versus-capture-time tradeoff (Mech-Mind parameters).
5. **Control projector brightness before electronic gain.** More projected photons improve shot-noise-limited SNR; camera gain amplifies noise as well as signal. Avoid heating or gamma/nonlinearity changes that invalidate calibration.
6. **Change geometry or surface optics when needed.** Cross-polarization, another view, matte scanning spray, or target rotation can suppress specular returns, but polarization also discards light and sprays alter the surface.

For a moving subject, HDR and additional phase steps create temporal misregistration. In that case, the best strategy may be a shorter single sequence at the highest non-clipping optical power, even if point completeness is lower.

4. Plane-based self-calibration without factory data

4.1 What is and is not identifiable

There are two materially different cases.

Case A: raw camera images and controllable projected patterns are available. Use a printed metric checkerboard/ChArUco/dot target on a flat board. Standard planar calibration estimates camera focal lengths, principal point, skew, radial/tangential distortion, and a pose for each view. Zhang's planar method requires the plane at several different orientations; nearly parallel poses are a degenerate configuration, with best conditioning in Zhang's simulation around 45 degrees before feature foreshortening becomes excessive. Treat the projector as an inverse camera: absolute phase supplies camera-to-projector pixel correspondences, then calibrate its intrinsics and the rigid camera-projector transform, as demonstrated by Li et al..

Case B: only the depth/range image or reconstructed cloud is available. Plane residuals can estimate range scale, offset, field-dependent bias, and some nonlinear warping. Herrera, Kannala, and Heikkila use planar surfaces from varied poses and additional full-frame walls at several depths to constrain a per-pixel disparity distortion model.

But several fronto-parallel unmarked planes at different distances do **not** robustly identify focal length and principal point. Changing focal length or principal point in a pinhole back-projection is an affine transformation of X and Y, and a fronto-parallel constant-Z plane remains planar. Translation and unknown plane pose absorb much of the change. Radial distortion may reveal itself as curvature and may weakly identify a distortion center when the board spans the field, but it is confounded with field-dependent depth bias; the lens principal point and radial-distortion center need not be exactly the same physical parameter. Known metric features, varied tilt around both axes, off-axis coverage, or an external pose/distance constraint are needed for a defensible full calibration.

Also establish whether the device reports axial Z, Euclidean range along a ray, disparity, phase, or already rectified/reconstructed XYZ. Applying a second optical distortion model to an already corrected point cloud can make it worse.

4.2 Recommended acquisition design

For a practical factory-independent calibration:

- Warm the unit to stable operating temperature and lock all auto settings for the calibration sequence.
- Use a genuinely flat, matte, dimensionally stable board with a measured fiducial pattern. Paper taped loosely to foam board is not a metric plane.
- Capture roughly 15-30 useful poses, not many copies of one pose: 3-5 distances spanning the intended volume; tilt about both axes from about 15 to 45 degrees; rotate in plane; and move the target so corners and edges of the image are covered.
- Include several full-frame matte wall/board captures at known distances to estimate smooth field-dependent depth bias separately from optical intrinsics.
- Retain raw projected images, phase, fringe order, amplitude/modulation, saturation masks, confidence, and output XYZ.
- Split data into calibration and held-out validation poses. Do not judge the model only by the residual it was optimized to minimize.

4.3 Model and objective

Let theta contain camera intrinsics (f_x, f_y, c_x, c_y), distortion coefficients (for example k_1, k_2, k_3, p_1, p_2), depth/disparity mapping parameters, camera-projector extrinsics, and any deliberately low-order field correction. For observation i in plane j:

1. Undistort pixel (u_i,v_i) with theta.
2. Convert raw phase/disparity q_i to metric depth or ray distance.
3. Back-project it to X_i(theta).

4 Fit plane pi_j = (n_j,d_j) and minimize a robust point-to-plane residual.

```
minimize sum rho((n_j dot X_i(theta) + d_j)^2 / sigma_i^2) + reprojection terms +
regularization
```

Use Huber/Tukey loss for gross outliers, but never let robust loss hide an entire coherent alias surface. Weight by a measured variance model, not by firmware confidence unless confidence has been calibrated. If fiducial corners and projector coordinates are available, joint camera/projector reprojection error carries far more intrinsic information than planarity alone.

Avoid an unconstrained per-pixel correction unless the application and data volume justify it. It can memorize board poses and erase real object curvature. A hierarchy is safer: physical camera-projector model first; low-order smooth residual field second; per-pixel table only with strong regularization, dense full-field coverage, and held-out tests.

4.4 Validation criteria

Report at least:

- plane RMS and robust 95th-percentile residual versus depth, image radius, and pose angle;
- signed mean error, not RMS alone, to expose scale/bias;
- residual heat maps in sensor coordinates to reveal fixed patterns;
- reconstructed known lengths and step heights at held-out positions;
- valid-point fraction and outlier fraction;
- parameter covariance/correlation or bootstrap stability, especially for c_x, c_y and high-order radial terms; and
- results before and after correction with identical filtering.

A low training-plane RMS does not prove correct principal point or correct 3D metric scale.

5. Noise models for triangulation depth sensors

5.1 Geometry and phase noise

For z = F/d, first-order propagation gives

```
sigma_z = (z^2/F) sigma_d
```

For a fringe pitch P and phase standard deviation sigma_phi,

```
sigma_d approximately equals (P/(2 pi)) sigma_phi
```

so a useful local model is

```
sigma_z approximately equals (z^2 P/(2 pi F)) sigma_phi
```

Khoshelham and Oude Elberink derive and experimentally verify the quadratic distance term for Kinect-style triangulation. The FPP-specific model of Lv and Qian then links camera intensity noise to phase and phase to 3D geometry. These equations describe **precision of valid, correctly decoded points**. They do not describe a wrong-order alias, specular multipath point, or threshold-induced missing point.

5.2 Distance, incidence angle, and albedo

Distance. Even if sigma_d or sigma_phi were constant, triangulation alone gives z-squared axial amplification. With fixed projector power and exposure, returned modulation often also falls with distance because projected irradiance spreads, optics vignette, and pattern focus changes. The total error can therefore rise faster than z squared. Auto-exposure can flatten part of the radiometric term while increasing motion sensitivity or clipping risk.

Albedo. Use reflectance at the projector wavelength, not visual RGB brightness. If direct modulation B scales approximately with rho E and shot noise dominates, the phase model implies sigma_phi proportional to 1/sqrt(rho E). If signal-independent read/quantization noise dominates, sigma_phi approaches 1/(rho E). With ambient light, A and shot noise increase without a matching increase in B. A dark target may show fewer returned points rather than a visibly larger standard deviation because the noisiest samples are rejected.

Incidence angle. Keep three angles separate:

- projector ray versus surface normal controls irradiance and projected fringe foreshortening;
- camera ray versus surface normal controls the observed footprint, occlusion, mixed pixels, and specular direction; and
- the camera-projector triangulation angle controls geometric leverage.

For an ideal Lambertian surface, projected irradiance contains a cosine-incidence factor and the camera sees approximately angle-independent radiance, but real surfaces add BRDF, polarization, shadowing, defocus, and microgeometry. At grazing angles, modulation usually falls, a camera pixel covers more surface, fringes compress or blur, and self-shadow/mixed-depth failures increase. NIST observed strong changes in returned point count and plane residual behavior as a flat plate was tilted, and warns that location and orientation change point density (Rachakonda et al.). There is no universal single cosine exponent that fits all structured-light cameras.

5.3 A practical statistical model

Use a mixture, not one Gaussian:

```
p(z_measured) = p_valid N(z_true + bias(u,v,z), sigma_z^2) + p_outlier H + p_alias sum_m w_m
delta(z - z_m) + p_missing
```

where H is a heavy-tailed or broad outlier distribution and z_m are the phase-order candidates. For valid points, one workable variance model is

```
sigma_z^2 = [z^2 P/(2 pi F)]^2 * 2(C_noise + K A)/(B^2 N) + sigma_floor^2
```

then add a smooth residual term for image position, target angle, temperature, or defocus only if held-out data support it. Do not absorb coherent bias into sigma.

The full 3D covariance is anisotropic. Lateral pixel quantization grows roughly with z/f, while disparity/phase error is amplified mainly along the triangulation/range direction and often dominates axial uncertainty at long range. For fusion or plane fitting, propagate measurement covariance through the calibrated ray-intersection Jacobian rather than attaching one isotropic radius to every point.

5.4 How to measure the model

Use matte panels with at least three known wavelength-specific reflectances and repeat scans at a grid of:

- distances across the full intended volume;
- surface tilts about both axes;
- exposure/projector-power settings; and
- image locations.

At each condition, collect enough repeated frames to separate temporal variance from fixed spatial bias. Fit a plane jointly or use a trusted external plane, then record signed bias, robust standard deviation, valid fraction, saturation fraction, modulation B, and alias/outlier fractions. First characterize fixed manual exposure; repeat later with the production auto-exposure logic as a separate system. Otherwise the controller hides the sensor's native dependence.

6. Recommended diagnostic campaign

Phase A: establish the alias law

1. Use one large matte gray plane normal to the central camera ray.
2. Move it through at least two suspected ambiguity intervals in small, surveyed steps.
3. At every step save raw phase, inferred fringe order/projector coordinate, modulation, and unfiltered depth.
4. Test whether bad depths satisfy 1/z_bad - 1/z_true = constant times integer.
5. If fringe pitch can change, repeat. Reciprocal-depth spacing should scale with P.

Phase B: isolate internal and scene light transport

1. Capture camera-dark, projector-off, projector-on with an external black absorber, and normal matte-board sequences.
2. Sweep projector power and exposure independently.
3. Translate and rotate a small bright patch across an otherwise dark board.
4. Inspect whether the suspicious phase/amplitude mask is fixed in camera coordinates, follows the patch, or follows the phase-order family.
5. Temporarily add external baffling or a removable polarizer if safe; never disturb factory geometry during a baseline run.

Phase C: decode the firmware controls

Run a one-factor sweep for background threshold, gradient threshold, and Z range with the same saved raw acquisition. Plot valid count, alias count, phantom-surface area, plane RMS, modulation distribution, and saturation fraction. A setting that merely hides points is not fixing the optical or correspondence error.

Phase D: calibrate only after classification

Do not fit a flexible calibration surface while aliases or ghosts remain. Otherwise the optimizer can encode the failure into a correction map that appears excellent on boards and fails on real geometry. Once gross failure modes are masked or resolved, calibrate the physical model, validate on held-out tilted planes and known steps, and only then fit a small residual field if necessary.

Material limitations and disagreements

- The firmware labels are generic and sometimes used inconsistently. The report gives conventional meanings and discriminating tests, not a claim about the undisclosed implementation.
- Published work directly documenting **housing-internal** projector-to-camera ghosts in commercial fringe sensors is sparse. The fixed-phantom explanation is a physics-based inference combining established optical ghost formation with established structured-light sensitivity to indirect illumination. It should be confirmed with raw-data tests.
- The simple alias and noise equations assume local rectification, pinhole geometry, and small errors. A calibrated general camera-projector Jacobian should replace them for quantitative prediction.
- Blue-light marketing often merges several effects. Narrow-band filtering and high source brightness have clear engineering value; a blanket claim that shorter wavelength alone removes reflections is not supported across all materials and geometries.
- Plane-only self-calibration is highly dependent on what raw observables exist. A depth map already corrected inside firmware does not expose the same parameters as raw camera/projector images.

Sources

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Research record

Searches covered fringe-phase unwrapping and geometric depth range; phase-order error geometry; structured-light global illumination and internal ghost optics; saturation and HDR methods; commercial exposure controls; planar camera/depth calibration and degeneracy; and triangulation/FPP precision models. Follow-up searches targeted the three firmware labels and the identifiability of principal point from parallel planes. Research stopped when all consequential claims had primary, government, or first-party support, and the remaining device-specific uncertainty could only be resolved by device documentation or experiment.